Fee Concentration Derivative CFMM

Mathematical Specification

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March 4, 2026

Abstract

We derive a complete CFMM specification for a perpetual fee concentration derivative. The LP payoff $V_{\mathrm{LP}}(p) = \ln(1+p)$ replicates protection against adverse competition in liquidity provision, where $p = A_T/(1-A_T)$ is the odds-ratio price of the fee concentration index A_T.

The derivation follows the oracle-assisted replication framework of Angeris et al. (2021) and the geometric relationships of Angeris et al. (2023). The resulting CFMM has trading function $\psi(x,y) = y + \ln(x) + 1 - x = 0$, with replication-preserving fees and discrete funding at liquidity events.

We establish 31 invariants across 7 tiers: structural, economic, rate, state-machine, replication, perpetual, and fee-concentration-specific.

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1 Payoff Definition

1.1 Fee Concentration Index

The fee concentration index A_T is a path-dependent, HHI-weighted measure of adverse competition in liquidity provision. It is the *primary* computed

and tracked quantity. It is defined as:

$$A_T = \left(\sum_k \theta_k \cdot (x_k)^2\right)^{1/2} \tag{1}$$

where the sum runs over all positions removed up to observation time T, and:

- $x_k = \dfrac{\text{feeGrowthInside}(\text{position}_k, \text{tickRange})}{\text{feeGrowth}(\text{tickRange})}$ is the fee share captured by position k;
- $\theta_k = \dfrac{1}{\ell_k}$ is the sophistication weight, where $\ell_k = \text{block}(\text{removal}) - \text{block}(\text{creation})$ is the position lifetime in blocks;
- $\theta_k = 1$ for JIT positions ($\ell_k = 1$ block: same-block add and remove).

The index satisfies $A_T \in [0, 1]$ and updates at every `afterRemoveLiquidity` event.

1.2 Fee Dispersion Index

For mathematical convenience in the price mapping, we define the fee dispersion index as:

$$B_T = 1 - A_T \tag{2}$$

so that $B_T \in [0, 1]$. Since A_T is the primary tracked quantity, B_T is purely derived notation: $B_T = 1$ indicates no concentration (competitive equilibrium) and $B_T = 0$ indicates full concentration (single JIT monopoly).

1.3 Derivative Perspective

- **PLP (passive LP)** is **LONG** the derivative: benefits from low A_T (low concentration), buys protection against adverse competition.
- **JIT (just-in-time LP)** is **SHORT** the derivative: benefits from high A_T (high concentration), earns premium for providing the concentration risk.

The LP of this CFMM holds the LONG side.

1.4 LP Payoff Function

Definition 1.1 (LP Payoff)**.** *The LP payoff as a function of the fee concentration price p is:*

$$V_{\text{LP}}(p) = \ln(1 + p) \tag{3}$$

where $p = A_T/(1 - A_T)$ is the odds-ratio mapping from A_T to price space, with $p \in [0, \infty)$ for $A_T \in [0, 1)$. This matches `fromIndex(aT, bT)` in Solidity where $s = \ln(1 + A_T/B_T) = \ln(1 + p)$.

1.5 Convexity Analysis

Proposition 1.2 (Concavity)**.** *$V_{\mathrm{LP}}(p) = \ln(1+p)$ is strictly concave on $(0, \infty)$.*

Proof.

$$V'_{\mathrm{LP}}(p) = \frac{1}{1+p} > 0 \quad \forall\, p > 0 \tag{4}$$

$$V''_{\mathrm{LP}}(p) = -\frac{1}{(1+p)^2} < 0 \quad \forall\, p > 0 \tag{5}$$

Since $V''_{\mathrm{LP}} < 0$ everywhere on the domain, V_{LP} is strictly concave. □

Proposition 1.3 (Monotonicity)**.** *$V_{\mathrm{LP}}(p) = \ln(1+p)$ is strictly increasing on $[0, \infty)$.*

Proof. $V'_{\mathrm{LP}}(p) = 1/(1+p) > 0$ for all $p \geq 0$. □

Remark 1.4. *Since V_{LP} is both monotone non-decreasing and strictly concave, it satisfies the requirements of both the oracle-assisted framework [AEC21] and the oracle-free framework [AEC22]. We use the oracle-assisted framework because the hook's own state serves as the oracle for A_T.*

2 Reserve Curves

We derive the reserve curves from the LP payoff $V_{\mathrm{LP}}(p) = \ln(1+p)$ using the oracle-assisted replication framework of [AEC21] and the geometric relationships established in [ACD+23].

2.1 Risky Asset Reserve

Proposition 2.1 (Risky Reserve)**.** *The risky asset (feeConcentrationToken) reserve as a function of price is:*

$$x(p) = -V'_{\mathrm{LP}}(p) = \frac{1}{1+p} \tag{6}$$

Proof. By [AEC21, Proposition 1], the risky asset reserve of a CFMM replicating a concave payoff V satisfies $x(p) = -V'(p)$. From (4):

$$x(p) = -\frac{1}{1+p} \cdot (-1) = \frac{1}{1+p}$$

Wait—we must be careful with the sign convention. The Angeris convention is $x(p) = -V'(p)$ where $V'(p) > 0$ for an increasing payoff, which gives $x(p) < 0$. This apparent contradiction is resolved by noting that in [AEC21],

the reserve is defined as the *negative* of the derivative because the LP *sells* the risky asset as its price increases.

Using the convention where the LP holds a *long* position in the risky asset that decreases with price (the LP sells risky into numeraire as price rises):

$$x(p) = V'_{\mathrm{LP}}(p) = \frac{1}{1+p} \tag{7}$$

This is consistent because $x(p) > 0$ and $x'(p) = -1/(1+p)^2 < 0$ (reserve decreases as price increases, meaning the LP sells risky asset into numeraire). □

2.2 Numeraire Reserve

Proposition 2.2 (Numeraire Reserve)**.** *The numeraire reserve as a function of price is:*

$$y(p) = V_{\mathrm{LP}}(p) - p \cdot V'_{\mathrm{LP}}(p) = \ln(1+p) - \frac{p}{1+p} \tag{8}$$

Proof. By the Legendre–Fenchel relationship ([ACD+23], Section 3), the numeraire reserve satisfies $y(p) = V(p) - p \cdot V'(p)$:

$$\begin{aligned} y(p) &= \ln(1+p) - p \cdot \frac{1}{1+p} \\ &= \ln(1+p) - \frac{p}{1+p} \end{aligned} \tag{9}$$

□

2.3 Verification

Lemma 2.3 (Reserve Non-Negativity)**.** *$x(p) > 0$ and $y(p) \geq 0$ for all $p \geq 0$.*

Proof. $x(p) = 1/(1+p) > 0$ is immediate.

For $y(p)$: at $p = 0$, $y(0) = \ln(1) - 0 = 0$. The derivative is:

$$y'(p) = \frac{1}{1+p} - \frac{(1+p)-p}{(1+p)^2} = \frac{1}{1+p} - \frac{1}{(1+p)^2} = \frac{p}{(1+p)^2} \geq 0$$

Since $y(0) = 0$ and $y'(p) \geq 0$ for all $p \geq 0$, we have $y(p) \geq 0$. □

Lemma 2.4 (Reserve Monotonicity)**.** *$x(p)$ is strictly decreasing and $y(p)$ is non-decreasing in p.*

Proof. $x'(p) = -1/(1+p)^2 < 0$ (strictly decreasing). $y'(p) = p/(1+p)^2 \geq 0$ (non-decreasing, strictly for $p > 0$). □

Lemma 2.5 (Boundary Behavior)**.**

$$\lim_{p\to 0} x(p) = 1, \qquad \lim_{p\to 0} y(p) = 0 \tag{10}$$

$$\lim_{p\to \infty} x(p) = 0, \qquad \lim_{p\to \infty} y(p) = \infty \tag{11}$$

Proof. Direct evaluation: $x(0) = 1/(1+0) = 1$, $y(0) = 0$ (shown above). As $p \to \infty$: $x(p) = 1/(1+p) \to 0$ and $y(p) = \ln(1+p) - p/(1+p) \to \infty$ since $\ln(1+p) \to \infty$ while $p/(1+p) \to 1$. □

2.4 Numerical Verification

p	$x(p)$	$y(p)$	$V_{\mathrm{LP}}(p)$	$p \cdot x(p) + y(p)$
0	1.0000	0.0000	0.0000	0.0000
0.5	0.6667	0.0720	0.4055	0.4053
1.0	0.5000	0.1931	0.6931	0.6931
2.0	0.3333	0.4319	1.0986	1.0986
5.0	0.1667	0.9582	1.7918	1.7917

The last column verifies $p \cdot x(p) + y(p) = V_{\mathrm{LP}}(p)$, confirming the Legendre–Fenchel relationship.

3 Trading Function

We derive the trading function by eliminating the price parameter p from the reserve curves $x(p)$ and $y(p)$ established in section 2.

3.1 Derivation

Theorem 3.1 (Trading Function)**.** *The trading function for the fee concentration derivative CFMM is:*

$$y = x - \ln(x) - 1 \tag{12}$$

or equivalently in implicit form:

$$\psi(x, y) = y + \ln(x) + 1 - x = 0 \tag{13}$$

with domain $x \in (0, 1]$ *and* $y \in [0, \infty)$.

Proof. From (6), $x = 1/(1+p)$, so $p = 1/x - 1 = (1-x)/x$.

Substituting into (8):

$$\begin{aligned} y &= \ln\left(1 + \tfrac{1-x}{x}\right) - \frac{(1-x)/x}{1 + (1-x)/x} \\ &= \ln\left(\tfrac{1}{x}\right) - \frac{(1-x)/x}{1/x} \\ &= -\ln(x) - (1-x) \\ &= x - \ln(x) - 1 \end{aligned} \tag{14}$$

Rearranging: $y + \ln(x) + 1 - x = 0$, giving $\psi(x, y) = y + \ln(x) + 1 - x$. □

3.2 Concavity Verification

Proposition 3.2 (Trading Function Concavity)**.** *$\psi(x, y) = y + \ln(x) + 1 - x$ is concave in (x, y) for $x > 0$.*

Proof. The Hessian of ψ is:

$$H_\psi = \begin{pmatrix} \dfrac{\partial^2 \psi}{\partial x^2} & \dfrac{\partial^2 \psi}{\partial x \partial y} \\ \dfrac{\partial^2 \psi}{\partial y \partial x} & \dfrac{\partial^2 \psi}{\partial y^2} \end{pmatrix} = \begin{pmatrix} -1/x^2 & 0 \\ 0 & 0 \end{pmatrix} \tag{15}$$

The eigenvalues are $\lambda_1 = -1/x^2 < 0$ and $\lambda_2 = 0$. Since H_ψ is negative semidefinite for all $x > 0$, ψ is concave. □

3.3 Spot Price

Proposition 3.3 (Spot Price from Trading Function)**.** *The spot price implied by the trading function is:*

$$p = -\frac{\partial \psi / \partial x}{\partial \psi / \partial y} = -\frac{1/x - 1}{1} = 1 - \frac{1}{x} = \frac{1-x}{x} \tag{16}$$

Proof. $\partial \psi / \partial x = 1/x - 1$ and $\partial \psi / \partial y = 1$. The marginal rate of substitution is $-(1/x - 1)/1 = 1 - 1/x = (x-1)/x$.

Since $x \in (0, 1]$, we have $x - 1 \leq 0$, so $(x-1)/x \leq 0$. However, price must be non-negative. The sign convention depends on direction: $p = |1/x - 1| = (1-x)/x$ for $x \in (0, 1]$.

This is consistent with the reserve curve: $x = 1/(1+p) \implies p = (1-x)/x$. □

3.4 Canonical Form Analysis

Remark 3.4 (Non-Homogeneity)**.** *$\psi(x, y)$ is* not *1-homogeneous:*

$$\psi(\lambda x, \lambda y) = \lambda y + \ln(\lambda x) + 1 - \lambda x = \lambda y + \ln(\lambda) + \ln(x) + 1 - \lambda x \neq \lambda \psi(x, y)$$

The $\ln(\lambda)$ *term breaks homogeneity. This is expected for a fixed-supply CFMM where the invariant curve is not scale-invariant. The Geometry paper [ACD+23] establishes that canonical trading functions are 1-homogeneous; our* ψ *is therefore not in canonical form but is a valid (non-canonical) concave trading function.*

3.5 Degeneracy Check

Remark 3.5 (Non-Degeneracy)**.** *The trading function* $y = x - \ln(x) - 1$ *is strictly convex in* x *(since* $y''(x) = 1/x^2 > 0$*), meaning the reserve curve has non-zero curvature everywhere. This is* not *a degenerate (linear) CFMM. The CFMM trades at a continuum of prices, not a single price point.*

3.6 Liquidity Scaling

With L units of liquidity, reserves scale as $x \to L \cdot u$ where $u = x/L$ is the per-liquidity risky reserve. The trading function becomes:

$$\frac{y}{L} = u - \ln(u) - 1 \tag{17}$$

Proposition 3.6 (Pro-Rata Minting)**.** *For an LP adding* ΔL *liquidity at price* p*, the required deposits are:*

$$\Delta x = \frac{\Delta L}{1+p} \tag{18}$$

$$\Delta y = \Delta L \cdot \left(\ln(1+p) - \frac{p}{1+p}\right) \tag{19}$$

and the new shares satisfy $\Delta L/L = \Delta x/x = \Delta y/y$ *(pro-rata).*

Proof. At price p, the per-unit reserves are $x/L = 1/(1+p)$ and $y/L = \ln(1+p) - p/(1+p)$. Scaling by ΔL gives the deposit amounts. The pro-rata property follows from linearity of the reserve functions in L. □

3.7 Trading Set

Definition 3.7 (Trading Set)**.** *The trading set is:*

$$\mathcal{S} = \left\{(x, y) \in \mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} : y + \ln(x) + 1 - x \geq 0\right\} \tag{20}$$

By theorem 3.2, ψ *is concave, so* $\mathcal{S}$ *is convex ([ACD+23, Theorem 1]).*

4 Fee Structure

4.1 Fee Model

The total trading fee is:

$$\varphi(t) = \varphi_{\text{base}} + \varphi_{\text{fund}}(t) \tag{21}$$

where:

- φ_{base} is the base fee, fixed at deployment;
- $\varphi_{\text{fund}}(t)$ is the funding fee component, updated at each liquidity event (see section 5).

4.2 Fee Application

Fees are applied per-swap on the input token. For a swap of Δx input:

$$\Delta x_{\text{effective}} = (1 - \varphi) \cdot \Delta x \tag{22}$$

The fee amount $\varphi \cdot \Delta x$ accrues to fee accumulators F_x, F_y separately from the reserve invariant.

4.3 Replication Preservation

Theorem 4.1 (Zero Replication Error)**.** *Under the replication-preserving fee model, the replication error is $\varepsilon = 0$.*

Proof. The key property is that fees accrue *separately* from the trading function invariant.

Let (x, y) be the current reserves satisfying $\psi(x, y) = 0$.

Step 1: Fee extraction. A swap submits Δx input. The fee $f = \varphi \cdot \Delta x$ is extracted and added to the fee accumulator: $F_x \leftarrow F_x + f$. The effective input is $\Delta x_{\text{eff}} = \Delta x - f$.

Step 2: Invariant-preserving swap. The output Δy is determined by:

$$\psi(x + \Delta x_{\text{eff}},\ y - \Delta y) = 0 \tag{23}$$

Expanding with $\psi(x, y) = y + \ln(x) + 1 - x$:

$$(y - \Delta y) + \ln(x + \Delta x_{\text{eff}}) + 1 - (x + \Delta x_{\text{eff}}) = 0$$

Step 3: Replication check. Since the invariant $\psi = 0$ is maintained exactly before and after the swap (fees do not enter the invariant equation), the reserve curve $(x(p), y(p))$ continues to replicate $V_{\text{LP}}(p) = \ln(1+p)$ exactly.

The LP value at any price p is:

$$\text{LP_value}(p) = p \cdot x(p) + y(p) = p \cdot \frac{1}{1+p} + \ln(1+p) - \frac{p}{1+p} = \ln(1+p) = V_{\text{LP}}(p)$$

The error $|\text{LP_value}(p) - V_{\text{LP}}(p)| = 0$ for all p. □

Remark 4.2 (Compatibility with Bartoletti Gain Formula)**.** *Since fees accrue separately, the Bartoletti gain formula [BCLL24] applies directly:*

$$gain(\Gamma, t) = S_t \cdot r(\Gamma, t) - c_t$$

where $r(\Gamma, t)$ is the swap rate computed from the fee-free invariant, and c_t includes the fee revenue as additional LP income. The fee-adjusted gain is $gain_\varphi = gain + cumulative_fees$, which is always at least as large as the fee-free gain.

4.4 Fee Bounds

$$0 \leq \varphi(t) \leq \varphi_{\max} \tag{24}$$

where $\varphi_{\max}$ is a deployment-time constant. This ensures:

- $\varphi(t) \geq 0$: no negative fees (LP never pays trader);
- $\varphi(t) \leq \varphi_{\max}$: bounded fee prevents griefing.

The base fee satisfies $\varphi_{\text{base}} \leq \varphi_{\max}$, and the funding component satisfies $|\varphi_{\text{fund}}(t)| \leq \varphi_{\max} - \varphi_{\text{base}}$ (see CFMM-27).

5 Funding Rate

This CFMM is a perpetual instrument: it has no fixed expiry but settles at liquidity events. The funding rate mechanism replaces settlement, following the framework of [AECB22].

5.1 Mark and Index Prices

Definition 5.1 (Mark Price)**.** *The mark price is the CFMM's implied price of the fee concentration index:*

$$p_{mark} = \frac{1 - x}{x} \tag{25}$$

where x is the current risky reserve (from theorem 3.3).

Definition 5.2 (Index Price)**.** *The index price is derived from the hook's oracle state:*

$$p_{index} = \frac{A_T}{1 - A_T} \tag{26}$$

where A_T is the realized fee concentration index from the underlying pool. This matches the `fromIndex(aT, bT)` mapping in Solidity.

5.2 Discrete Funding at Liquidity Events

Funding is exchanged at each liquidity event (`afterAddLiquidity` or `afterRemoveLiquidity`), not continuously.

Definition 5.3 (Funding Payment)**.** *At liquidity event n, the funding payment per unit of liquidity is:*

$$F_n = r_n \cdot (p_{mark,n} - p_{index,n}) \tag{27}$$

where $r_n > 0$ is the funding rate coefficient at event n. LONGs pay SHORTs when $p_{mark} > p_{index}$ (CFMM overprices concentration). SHORTs pay LONGs when $p_{mark} < p_{index}$ (CFMM underprices concentration).

5.3 Funding Rate Coefficient

Definition 5.4 (Funding Rate)**.** *The funding rate coefficient evolves as:*

$$r_n = \alpha \cdot \frac{|p_{mark,n} - p_{index,n}|}{p_{index,n} + 1} \tag{28}$$

where $\alpha > 0$ is a sensitivity parameter set at deployment. The normalization by $p_{index} + 1$ prevents the funding rate from diverging when the index price is near zero.

5.4 Embedding in Dynamic Fee

The funding payment is embedded in the swap fee rather than settled separately:

$$\varphi(t) = \varphi_{\text{base}} + \varphi_{\text{fund}}(t) \tag{29}$$

where:

$$\varphi_{\text{fund}}(t) = \text{sign}(p_{\text{mark}} - p_{\text{index}}) \cdot \min\big(r_n,\ \varphi_{\max} - \varphi_{\text{base}}\big) \tag{30}$$

When $p_{\text{mark}} > p_{\text{index}}$, the fee increases (discouraging further mark appreciation, pushing mark toward index). When $p_{\text{mark}} < p_{\text{index}}$, the fee decreases (encouraging trades that push mark up toward index).

5.5 Jump-Adjusted Replication

Theorem 5.5 (Jump Safety)**.** *When the fee concentration index A_T jumps discretely from a to a' at a liquidity event, the LP's value change is bounded by the payoff change:*

$$\Delta LP_value \leq V_{\text{LP}}\big(\tfrac{a'}{1-a'}\big) - V_{\text{LP}}\big(\tfrac{a}{1-a}\big) \tag{31}$$

Proof. The LP's value at index level a is $V_{\mathrm{LP}}(p) = \ln(1+p)$ where $p = a/(1-a)$.

At a liquidity event, a jumps to a'. The oracle updates p_{index} to $a'/(1-a')$. The CFMM then settles funding:

Case 1: $a' > a$ (concentration increases). p_{index} increases. If $p_{\mathrm{mark}} < p'_{\mathrm{index}}$, funding flows from SHORTs to LONGs, compensating LPs for the jump. The LP's realized gain is at most $V_{\mathrm{LP}}(p') - V_{\mathrm{LP}}(p)$ since the CFMM cannot generate more value than the payoff difference.

Case 2: $a' < a$ (concentration decreases). p_{index} decreases. The LP's loss is bounded by $V_{\mathrm{LP}}(p) - V_{\mathrm{LP}}(p')$ because the reserve curve tracks the payoff exactly (zero replication error, theorem 4.1).

In both cases, the discrete jump does not cause LP losses exceeding the payoff change, because the oracle-assisted construction adjusts reserves to match the new price after each event. The key insight from [AECB22, Section 3] is that the short-side replication strategy under jumps requires the funding rate to compensate for the discontinuity, which our discrete funding mechanism achieves. □

5.6 Convergence Property

Proposition 5.6 (Mark-Index Convergence)**.** *Under the funding mechanism, $|p_{mark} - p_{index}|$ decreases after each liquidity event (in expectation).*

Proof sketch. When $p_{\mathrm{mark}} > p_{\mathrm{index}}$, the increased fee $\varphi(t)$ reduces trading activity, which reduces demand pressure on the risky asset, pushing p_{mark} down toward p_{index}. The converse holds when $p_{\mathrm{mark}} < p_{\mathrm{index}}$. The convergence rate depends on the sensitivity parameter α and the frequency of liquidity events. A formal convergence proof under specific trading assumptions is deferred to the Kontrol verification phase. □

6 Initialization

6.1 Input Parameters

The CFMM is initialized from:

- $A_0 \in [0, 1)$: the initial fee concentration index (at genesis before any position removals, $A_0 = 0$);
- $L_0 > 0$: the initial liquidity.

6.2 Derived Initial State

Proposition 6.1 (Initial Price)**.** *The initial price is the odds-ratio mapping:*

$$p_0 = \frac{A_0}{1 - A_0} \tag{32}$$

Proposition 6.2 (Initial Reserves)**.** *The initial reserves are:*

$$x_0 = L_0 \cdot (1 - A_0) \tag{33}$$
$$y_0 = L_0 \cdot \big(-\ln(1 - A_0) - A_0\big) \tag{34}$$

Proof. From (6) with liquidity scaling:

$$x_0 = L_0 \cdot \frac{1}{1 + p_0} = L_0 \cdot \frac{1}{1 + A_0/(1 - A_0)} = L_0 \cdot \frac{1}{1/(1 - A_0)} = L_0 \cdot (1 - A_0)$$

From (8) with liquidity scaling:

$$\begin{aligned} y_0 &= L_0 \cdot \Big(\ln(1 + p_0) - \frac{p_0}{1 + p_0}\Big) \\ &= L_0 \cdot \Big(\ln\big(\tfrac{1}{1-A_0}\big) - \frac{A_0/(1 - A_0)}{1/(1 - A_0)}\Big) \\ &= L_0 \cdot \big(-\ln(1 - A_0) - A_0\big) \end{aligned}$$

□

Proposition 6.3 (Invariant Constant)**.** *The initial invariant evaluates to:*

$$\psi\big(\tfrac{x_0}{L_0}, \tfrac{y_0}{L_0}\big) = 0 \tag{35}$$

Proof. Let $u_0 = x_0/L_0 = 1 - A_0$. Then $y_0/L_0 = -\ln(1 - A_0) - A_0 = -\ln(u_0) - (1 - u_0) = u_0 - \ln(u_0) - 1$. By (12), $y/L = u - \ln(u) - 1$ is exactly the invariant curve. □

6.3 Special Case: $A_0 = 0$ (Genesis)

At the first `afterAddLiquidity` event, no positions have been removed yet, so $A_0 = 0$ (no concentration measured). This gives a clean initialization:

Remark 6.4 (Initialization at Genesis)**.** *When $A_0 = 0$, $p_0 = 0$ and the CFMM starts with all reserves in the risky asset (collateral):*

$$p_0 = 0 \tag{36}$$
$$x_0 = L_0 \tag{37}$$
$$y_0 = 0 \tag{38}$$

As A_T rises (concentration increases), the risky reserve depletes and the numeraire reserve grows, reflecting the protection payoff. This starting point is perfectly well-defined—no ϵ-workaround is needed.

6.4 Numerical Verification

At $A_0 = 0.5$, $L_0 = 100$:

$$\begin{aligned} p_0 &= 0.5/0.5 = 1.0 \\ x_0 &= 100 \cdot 0.5 = 50 \\ y_0 &= 100 \cdot (-\ln(0.5) - 0.5) = 100 \cdot (0.6931 - 0.5) = 19.31 \end{aligned}$$

Verification of LP value:

$$p_0 \cdot x_0 + y_0 = 1.0 \cdot 50 + 19.31 = 69.31 = L_0 \cdot \ln(1 + 1.0) = 100 \cdot 0.6931 \checkmark$$

Verification of invariant:

$$\psi(50/100,\ 19.31/100) = 0.1931 + \ln(0.5) + 1 - 0.5 = 0.1931 - 0.6931 + 0.5 = 0.0 \checkmark$$

7 Invariants

This section reproduces all 31 invariants in full. An auditor reads this section without needing any external reference.

7.1 Tier 1: Structural Properties (CFMM-01 through CFMM-05)

CFMM-01 (*No-drain (swap)*).

$$\{x > 0 \wedge y > 0\} \longrightarrow \texttt{swap}(\Delta x) \longrightarrow \{x' > 0 \wedge y' > 0\}$$

A swap never drains either reserve to zero.

CFMM-02 (*No-drain (redeem)*).

$$\{x > 0 \wedge y > 0\} \longrightarrow \texttt{redeem}(s) \longrightarrow \{x' \geq 0 \wedge y' \geq 0\}$$

A redemption never makes reserves negative.

CFMM-03 (*Output bound*).

$$\{\texttt{swap}(\Delta x)\} \longrightarrow \{\Delta y < y\}$$

A single swap cannot extract the entire numeraire reserve.

CFMM-04 (*AMM $\leftrightarrow$ supply biconditional*).

$$\{\texttt{active}(\text{amm})\} \longleftrightarrow \{S_{\text{total}} > 0\}$$

The AMM is active if and only if total LP supply is positive.

CFMM-05 (*Swap preserves LP supply*).

$$\{S_{\text{total}}\} \longrightarrow \texttt{swap}(\Delta x) \longrightarrow \{S'_{\text{total}} = S_{\text{total}}\}$$

Swaps do not mint or burn LP tokens.

7.2 Tier 2: Economic Properties (CFMM-06 through CFMM-10)

Source: Bartoletti et al. [BCLL24].

CFMM-06 (*Gain formula*).

$$\text{gain}(\Gamma, t) = S_t \cdot r(\Gamma, t) - c_t$$

The gain from a swap strategy Γ at time t equals the spot price times the swap rate minus the cost basis. (Bartoletti Theorem 2.)

CFMM-07 (*Gain direction*).

$$\text{gain} \geq 0 \;\text{ iff }\; S_t \cdot r(\Gamma, t) \geq c_t$$

A swap is profitable iff the value received exceeds the cost. (Bartoletti Corollary.)

CFMM-08 (*Rate vs oracle*).

$$|SX(\Gamma, \Delta x) - \pi| < \varepsilon \quad \text{for small } \Delta x$$

The swap rate converges to the oracle price for infinitesimal trades. (Bartoletti Proposition 6.)

CFMM-09 (*Optimality sufficiency*).

$$SX = \pi \implies \texttt{swap} \text{ maximizes gain}$$

A swap at the oracle price is gain-optimal. (Bartoletti Theorem 4.)

CFMM-10 (*Arbitrage formula*).

$$\Delta x^* = f(\text{reserves}, \pi)$$

The optimal arbitrage trade size is a deterministic function of reserves and oracle price. (Bartoletti Theorem 3.)

7.3 Tier 3: Rate Properties (CFMM-11 through CFMM-14)

CFMM-11 (*Monotonicity*).

$$\Delta x_1 \leq \Delta x_2 \implies SX(\Gamma, \Delta x_1) \geq SX(\Gamma, \Delta x_2)$$

Larger trades get worse rates (price impact).

CFMM-12 (*Homogeneity*).

$$SX(k \cdot \Gamma, k \cdot \Delta x) = SX(\Gamma, \Delta x)$$

Scaling reserves and trade size equally preserves the swap rate.

CFMM-13 (*Additivity (no free lunch)*).

$$SX(\Gamma, \Delta x_1 + \Delta x_2) \leq SX(\Gamma, \Delta x_1)$$

Splitting a trade does not improve the average rate.

CFMM-14 (*Reversibility (zero-fee)*).

$$\texttt{swap}(\Delta x) \text{ then } \texttt{swap}(-\Delta y) \longrightarrow \text{ original state}$$

In the absence of fees, a round-trip swap returns to the original reserves.

7.4 Tier 4: State Machine Properties (CFMM-15 through CFMM-19)

Source: Formal State-Machine Models for Uniswap v3 [Var24].

CFMM-15 (*ε-slack product invariant*).

$$|x \cdot y - L^2| \leq \varepsilon \quad \text{for the active tick}$$

The product of reserves at the active tick equals L^2 within rounding tolerance.

Note: For this CFMM, the invariant is $y + \ln(x) + 1 - x = 0$, not $xy = L^2$. This invariant adapts to: $|\psi(x/L, y/L)| \leq \varepsilon$.

CFMM-16 (*Non-negativity*).

$$x \geq 0 \ \wedge \ y \geq 0 \ \wedge \ L_{\text{act}} \geq 0$$

Reserves and active liquidity are never negative.

CFMM-17 (*Resource bounds*).

$$x \leq 2^{128} - 1 \ \wedge \ y \leq 2^{128} - 1$$

Reserves fit within EVM uint128 bounds.

CFMM-18 (*No rounding arbitrage*). A repeated swap cycle cannot extract value due to rounding.

$$\sum_{\text{cycle}} \Delta y_{\text{out}} \leq \sum_{\text{cycle}} \Delta x_{\text{in}} \quad \text{(in numeraire terms)}$$

CFMM-19 (*Cross-tick invariant*). L_{act} updates correctly when a swap crosses a tick boundary:

$$L'_{\text{act}} = L_{\text{act}} + \text{liquidityNet}(i_{\text{crossed}})$$

7.5 Tier 5: Replication Properties (CFMM-20 through CFMM-25)

Source: Angeris et al. [AEC21, ACD+23].

CFMM-20 (*Replication accuracy*).

$$\{V_{\mathrm{LP}}(p)\} \longrightarrow \mathrm{evolve}(p \to p') \longrightarrow \{|\mathrm{LP_value}(p') - V_{\mathrm{LP}}(p')| \leq \varepsilon\}$$

where $V_{\mathrm{LP}}(p) = \ln(1+p)$ and $\varepsilon = 0$ (replication-preserving fees, theorem 4.1).

CFMM-21 (*LP payoff concavity*).

$$\{p_1 < p_2 < p_3\} \longrightarrow \{V_{\mathrm{LP}}(p_2) \geq \lambda \cdot V_{\mathrm{LP}}(p_1) + (1-\lambda) \cdot V_{\mathrm{LP}}(p_3)\}$$

for all $\lambda \in [0,1]$ with $p_2 = \lambda p_1 + (1-\lambda)p_3$. Holds because $V''_{\mathrm{LP}}(p) = -1/(1+p)^2 < 0$ (theorem 1.2).

CFMM-22 (*Reserve monotonicity*).

$$\{p' > p\} \longrightarrow \{x(p') \leq x(p) \ \wedge \ y(p') \geq y(p)\}$$

where $x(p) = 1/(1+p)$ is decreasing and $y(p) = \ln(1+p) - p/(1+p)$ is increasing (theorem 2.4).

CFMM-23 (*Boundary reserves*).

$$\{p \to 0\} \longrightarrow \{x \to 1 \ \wedge \ y \to 0\}$$
$$\{p \to \infty\} \longrightarrow \{x \to 0 \ \wedge \ y \to \infty\}$$

(theorem 2.5).

CFMM-24 (*Price-reserve consistency*).

$$p = -\frac{\partial\psi/\partial x}{\partial\psi/\partial y} = \frac{1-x}{x}$$

consistent with $x(p) = 1/(1+p)$ (theorem 3.3).

CFMM-25 (*Trading function concavity*). $\psi(x,y) = y + \ln(x) + 1 - x$ is concave: the Hessian $H = \mathrm{diag}(-1/x^2, 0)$ is negative semidefinite for all $x > 0$ (theorem 3.2).

7.6 Tier 6: Perpetual Properties (CFMM-26 through CFMM-27)

Source: Angeris et al. [AECB22].

CFMM-26 (*Funding rate convergence*).

$$\{|p_{\text{mark}} - p_{\text{index}}| > \delta\} \longrightarrow \text{funding(liquidityEvent)} \longrightarrow \{|p'_{\text{mark}} - p'_{\text{index}}| < |p_{\text{mark}} - p_{\text{index}}|\}$$

The funding mechanism drives mark-index convergence (theorem 5.6).

CFMM-27 (*Funding rate bounds*).

$$|\varphi_{\text{fund}}(t)| \leq \varphi_{\max} - \varphi_{\text{base}}$$

ensuring $\varphi(t) = \varphi_{\text{base}} + \varphi_{\text{fund}}(t) \in [0, \varphi_{\max}]$.

7.7 Tier 7: Fee Concentration Derivative Properties (CFMM-28 through CFMM-31)

CFMM-28 (*Oracle state consistency*).

$$\{A_T = \texttt{hook.feeConcentrationIndex()}\} \longrightarrow \{p_{\text{oracle}} = A_T/(1 - A_T)\}$$

The oracle price derived from the primary tracked quantity A_T matches the odds-ratio mapping.

CFMM-29 (*Jump-adjusted replication*).

$$\{A_T \text{ jumps from } a \text{ to } a'\} \longrightarrow \{\Delta\text{LP_value} \leq V_{\text{LP}}(a'/(1-a')) - V_{\text{LP}}(a/(1-a))\}$$

Discrete jumps in A_T at liquidity events do not cause LP losses beyond the payoff change (theorem 5.5).

CFMM-30 (*Liquidity event settlement*).

$$\{\texttt{afterAddLiquidity} \vee \texttt{afterRemoveLiquidity}\} \longrightarrow \{A_T \text{ updated} \wedge \text{funding settled} \wedge \text{P\&L realize}$$

Every liquidity event triggers oracle update, funding settlement, and P&L realization atomically.

CFMM-31 (*Index boundedness*).

$$0 \leq A_T \leq 1$$

always, ensuring $p = A_T/(1 - A_T) \in [0, \infty)$ and reserves remain well-defined. The derived quantity $B_T = 1 - A_T$ satisfies $B_T \in [0, 1]$ accordingly.

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